

1. open your desktop / laptop, locate the RAM slots and remove and reinsert the RAM module to simulate reseating then document each step you took and any issues you observed during the process.



1. power down completely
2. open the case & ground yourself
3. Release the RAM clips
4. pull RAM out
5. Re-insert RAM
6. Close & test

* observed issues :

- computer won't turn on or beeps :-

The stick isn't pushed down proper. It takes more force you might think to click the clip locked.

- won't fit in the slot :-

The stick is facing the wrong direction. flip it 180 degrees so the notch lines up.

2. create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.

→ 1. check cables :

check your monitor cable is plugged directly into the graphic card not the motherboard. Try a different cable.

2. Reseat the card :

unplug PC, push GPU power cable tight, and check card is snapped securely into its slot.

3. Reinstall Drivers :

Uninstall old graphic drivers then download and install newest drivers from NVIDIA/AMD.

4. Stress Test :-

Run a heavy game for 15 minutes, watch to see if it crashes or gets too hot.

5. Swap & test :-

Try GPU in another person's PC. If it fails there too, the card is dead. If it works, another part of your PC is the problem.

3. simulate a storage issue by unplugging and replugging a SATA or NVMe (Utop/SSD) drive on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

→ * DRIVE unplugged :-

* BIOS / UEFI screen :-

The drive list shows "No device found" or leaves the slot blank.

* Boot Error screen :-

You see black screen with message like ---

- "Reboot and select proper Boot device"
- "No Bootable device found"
- Automatically boots straight to BIOS setup instead of windows.

* After Reconnecting (drive plug Back) :-

* BIOS / UEFI screen :-

The drive's brand and size pop back up in storage list and boot menu.

* windows screen :-

Window boots normally. If it was a secondary storage drive, it instantly appears under this PC in File Explorer without any error message.

4. Given a scenario where a PC randomly restarts list which component you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

→ check RAM :-

Bad RAM is the 1st cause of sudden random PC restarts without warning. Storage usually freeze the system or throw boot errors. While CPU issues usually crash to desktop or mess up the visuals before shutting off.

* Diagnostic steps for RAM :-

1. Reseat the sticks :-

Turn off PC, pop RAM out, blow any dust and push it back.

2. Run one stick at a time :-

If have multiple RAM sticks, take all out except one. Turn on PC & test it. Swap sticks find the bad one.

3. Run memtest86 :-

Download free tool memtest86 to USB drive. Boot into it, let it test your RAM for errors. Any red errors mean the stick is bad and need replacing.